

Question	Marking Guidance	Mark	Comments
6.1	1. Concentration of mineral ion/named mineral ion in soil; 2. Soil pH; 3. Temperature; 4. Light intensity/wavelength/duration; 5. Distance between seeds/plants; 6. Volume of water given; 7. CO₂ concentration; 8. Humidity;	2 max	1 and 2. Allow 'growing solution' for 'soil'. 2. pH alone is insufficient. 3. Allow 'colour of light' Reject 'amount' for mps 1, 4, 6 and 7. Ignore O₂ concentration Three correct = 2 marks Two correct = 1 mark One or none correct = 0 marks

6.2	2,4-D causes an increase in release of ions from wild oat cells and 2,4-D does not affect/has little effect on the release of ions from wheat cells; (For wheat) Difference is less than LSD/7 so <u>difference</u> is not significant; OR (For wild oats) Difference is more than LSD/10 so <u>difference</u> is significant; - Loss of ions from cells (likely to) lead to cell/plant death/damage; OR Disruption of cell membrane (likely to) lead to cell/plant death/damage; - No evidence here about death of plants as a result of this ion loss; - No evidence here of other ecological/environmental impact;	4 max	- Accept reference to 'concentration of ions in water' or 'disruption of the cell membranes' in place of 'release of ions' 1. Accept 'difference in release of ions from wild oats is 25 and difference in release of ions from wheat is 1' - Accept '(For wheat) difference is less than LSD so greater than 5% probability that <u>difference</u> is due to chance' OR '(For oats) difference is more than LSD so less than 5% probability that <u>difference</u> is due to chance' - Accept 'development of resistance'
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6.3	1. (Maintain temperature) so that the rate of diffusion (of ions out of cells) remains constant OR (Maintain temperature) so no change in fluidity of phospholipids/kinetic energy of phospholipids; OR (Maintain temperature) so no change in shape/structure/denaturation of membrane proteins; 2. (Shaking) So all surfaces of the leaf discs are exposed (to water)/so all submerged; OR To maintain diffusion/concentration gradient (for ions out of leaf discs);	2	1. Ignore references to rate of enzyme catalysed reactions 2. Accept 'so that leaf discs do not stick together'
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